Math 5324H Lec16

Suppose the wave equation

IVP $\begin{cases} u_{tt} - c^2 u_{xx} = 0, & x \in \mathbb{R},\ t > 0 \\ u(x,0) = \phi(x), & x \in \mathbb{R} \\ u_t(x,0) = \psi(x), & x \in \mathbb{R} \end{cases}$

Energy: $E[u] = \frac{1}{2}\int_{\mathbb{R}} |u_t|^2 + |u_x|^2 \, dx$

We have already prove $\frac{dE}{dt} = 0 \Rightarrow \underline{E[u](t) = E[u](0)}$

Uniqueness: u_1 and u_2 satified IVP

$w = u_1 - u_2$

$\begin{cases} w_{tt} - c^2 w_{xx} = 0, & x \in \mathbb{R},\ t > 0 \\ w(x,0) = \phi_1 - \phi_2 = 0, & x \in \mathbb{R} \\ w_t(x,0) = \psi_1 - \psi_2 = 0, & x \in \mathbb{R} \end{cases}$

$\underline{0 \le E[w](t) = E[w](0)}$

$E[u](0) = \frac{1}{2}\int_{\mathbb{R}} |\underset{0}{w_t}|^2 + |\underset{0}{w_x}|^2 \, dx$

$\underline{0 \le E[w](t)} = \underline{E[w](0) = 0}$

$\Downarrow$

$w_t(x,t) = 0 \Rightarrow \cancel{w(x,t) = x + C}$

$w_x(x,t) = 0 \Rightarrow \cancel{w(x,t) = t + C}$

$w(x,t)$ is constant ~~in~~ time

$w(x,t)$ is constant in x

$\boxed{w(x,t) = \text{constant}}$ $\xrightarrow{w(x,0) = 0}$ $\boxed{w(x,t) = 0}$

$u_1 = u_2 \quad \forall (x,t)$

Property:

① ~~Sa~~ Scaling $\lambda > 0$

$u = (\sqrt{\lambda}x, \lambda t)$ is also solution

~~② $u(-x,t)$ is also solution~~

② $u(x, t-h)$ is also solution

Translation invariance ⇒ ✓ ③ $u(x-y, t)$ is also solution

✓ ④ Any derivative of u is also solution

卷积解构 ⇒ ✓ ④ $\int_R u(x-y,t)\, g(y)\, dy$ is also solution

把平移后的解按权重 $g(y)$ 叠加

这个解 一个只依赖 y 的函数（权重函数）

五、Heaviside 阶跃函数 ⇒ $H(x) = \begin{cases} 1 & x \geq 0 \\ 0 & x < 0 \end{cases}$ $H(\lambda x) = H(x)$

$Q(x,0) = \begin{cases} 1 & x \geq 0 \\ 0 & x < 0 \end{cases}$ ⇐ 温度分布是1，温度分布

物理图像：一个无限长的棒子，左边始终温度为0，右边为1，然后开始热扩散

$$\begin{cases} Q_t - kQ_{xx} = 0, & x \in R,\ t > 0 \\ Q(x,0) = H(x), & x \in R \end{cases}$$

这边和两个常数，所以为 (x,t)

$(\sqrt{\lambda}x, \lambda t)$ 解仍不变

② 初值 $H(x)$ 仍解仍不变

$Q(\sqrt{\lambda}x, \lambda t)$ is also solution

$Q(x,t) = q\left(\frac{x}{\sqrt{t}}\right)$ self-semitor variable 自相似变量

$q(\sqrt{k}z) = q(\sqrt{k}z)$ 相似变量

$= q\left(\frac{x}{\sqrt{4kt}}\right)$ similarity variable 相似变量

① 空间的缩放 $x \sim \sqrt{t}$

② 水泥的标准差

$= q(p)$ $p: \frac{x}{\sqrt{4kt}}$

$$P = \frac{x}{\sqrt{4kt}} = x(4kt)^{-\frac{1}{2}}, \quad P_t = x\cdot\left(-\frac{1}{2}\right)(4k)(4kt)^{-\frac{3}{2}}$$

$$Q_t = -\frac{1}{2t}\frac{x}{\sqrt{4kt}}g'(p)$$

$$Q_x = \frac{1}{\sqrt{4kt}}g'(p)$$

$$Q_{xx} = \frac{1}{4kt}g''(p)$$

$$0 = Q_t - kQ_{xx} = \frac{1}{t}\left[-\frac{1}{2}pg'(p) - \frac{1}{4}g''(p)\right]$$

$$\Downarrow$$

$$\boxed{g'' + 2pg' = 0} \quad (g = g(p))$$

$$\boxed{\int_0^{+\infty} e^{-s^2}ds = \frac{\sqrt{\pi}}{2}}$$

$$Q(x,t) = \boxed{g(p) = C_1\int e^{-p^2}dp + C_2}$$

$$= C_1\int^{x/\sqrt{4kt}} e^{-p^2}dp + C_2 \quad (t>0)$$

if $x>0$ $\quad 1 = \lim_{t\to 0^+} Q(x,t) = C_1\int_0^{\infty} e^{-p^2}dp + C_2$

$$= C_1\frac{\sqrt{\pi}}{2} + C_2$$

if $x<0$, $\quad 0 = \lim_{t\to 0^+} Q(x,t) = C_1\int_0^{-\infty} e^{-p^2}dp + C_2$

$$= -C_1\frac{\sqrt{\pi}}{2} + C_2$$

$$\Downarrow$$

$$\begin{cases} C_2 = \frac{1}{2} \\ C_1\sqrt{\pi} = 1 \Rightarrow C_1 = \frac{1}{\sqrt{\pi}} \end{cases}$$

$$Q(x,t) = \frac{1}{2} + \frac{1}{\sqrt{\pi}}\int^{x/\sqrt{4kt}} e^{-p^2}dp \quad (t>0)$$